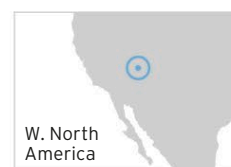




Bryce Canyon

A natural amphitheater filled with thousands of strange stone spires, columns, and towers—more than anywhere else on Earth



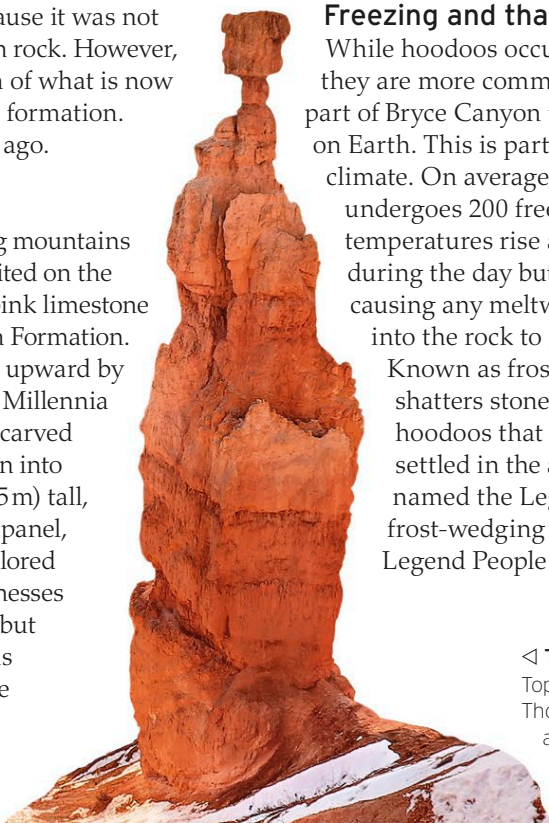
△ BRYCE AMPHITHEATER

The rim of Bryce Canyon is 8,000–9,000 ft (2,400–2,700 m) high—perfect for overlooking the national park’s “amphitheater” area.

Bryce is not a conventional canyon because it was not created by flowing water cutting through rock. However, an ancient lake that once covered much of what is now southwestern Utah did play a role in its formation. The process began 55–35 million years ago.

A sea of stone totems

Mineral-rich sediment from surrounding mountains was washed down by rainfall and deposited on the lake’s base, eventually turning into the pink limestone that geologists today know as the Claron Formation. Geological upheaval forced the lakebed upward by nearly 1 mile (1.6 km) to create a plateau. Millennia of weathering and erosion subsequently carved the Claron Formation first into cliffs, then into tall columns of rock, some a mere 5 ft (1.5 m) tall, others rising as high as 150 ft (45 m) (see panel, right). Known as hoodoos, these multicolored towers contain sections of varying thicknesses and are composed mainly of limestone but with silt and mudstone as well. Minerals such as iron oxide and manganese oxide add pink, purple, and blue tints. The different hues and lumpy, unequal shapes of the stone spires remind some visitors of wooden totem poles.



Freezing and thawing

While hoodoos occur on every continent, they are more common in the northern part of Bryce Canyon than anywhere else on Earth. This is partly due to the region’s climate. On average, Bryce Canyon undergoes 200 freeze-thaw cycles a year; temperatures rise above freezing point during the day but drop below it at night, causing any meltwater that has seeped into the rock to freeze and expand.

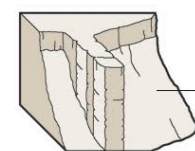
Known as frost-wedging, this process shatters stone, reshaping it into the hoodoos that the Paiute Indians, who settled in the area around 1200 CE, named the Legend People. As frost-wedging continues, many Legend People collapse each year.

◁ THOR’S HAMMER

Top-heavy hoodoos, such as Thor’s Hammer, are fragile and at greatest risk of collapse, especially during spring and fall, when frost-wedging is at its peak.

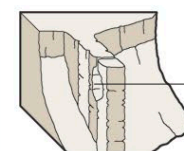
HOW A HOODOO FORMS

Rainwater washes debris away from the plateau’s edges, forming fins. Frost-wedging creates ice windows in the fins, separating them into hoodoos. Rain then rounds off the edges, lending a lumpy appearance.



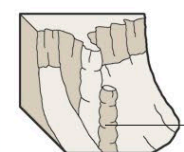
rainwater erodes slopes, carrying away debris

FIN



frost-wedging creates ice window in fin

WINDOW



section separates from fin, forming hoodoo

HOODOOS